

Mischievous Little Guy

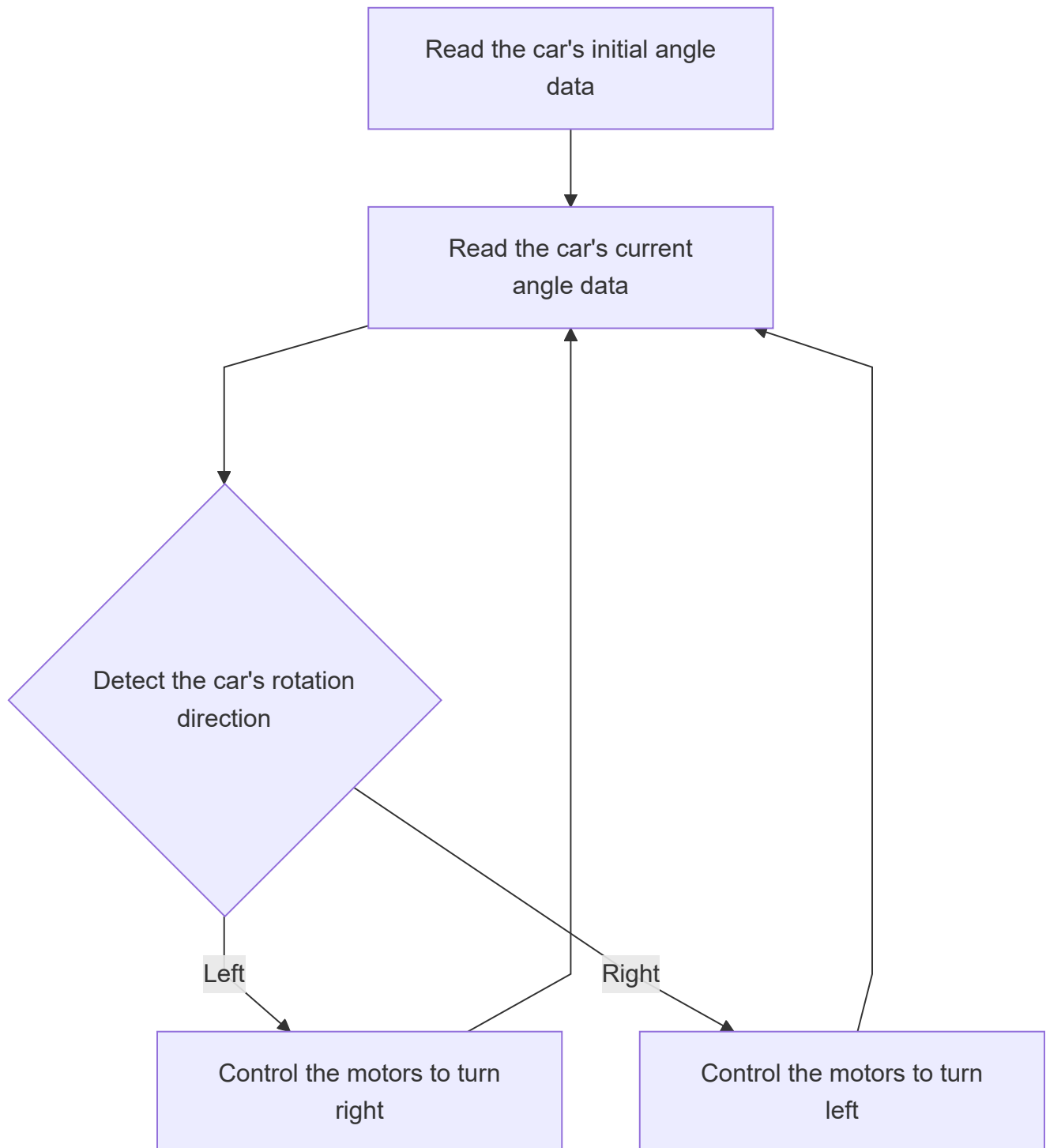
Mischievous Little Guy

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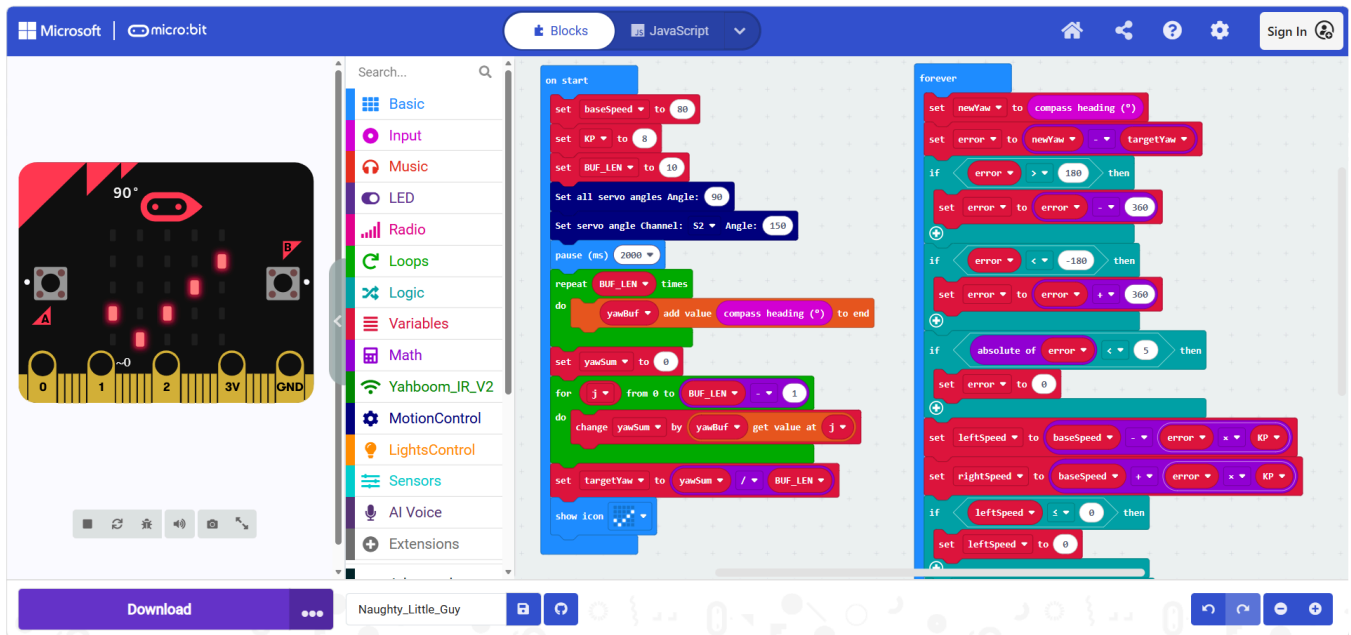
1. Experiment Purpose

The digital compass is an input sensor that detects magnetic fields. The micro:bit has a built-in compass that can detect direction. In this section, we will combine it with motor control to achieve the "Mischievous little guy" effect.

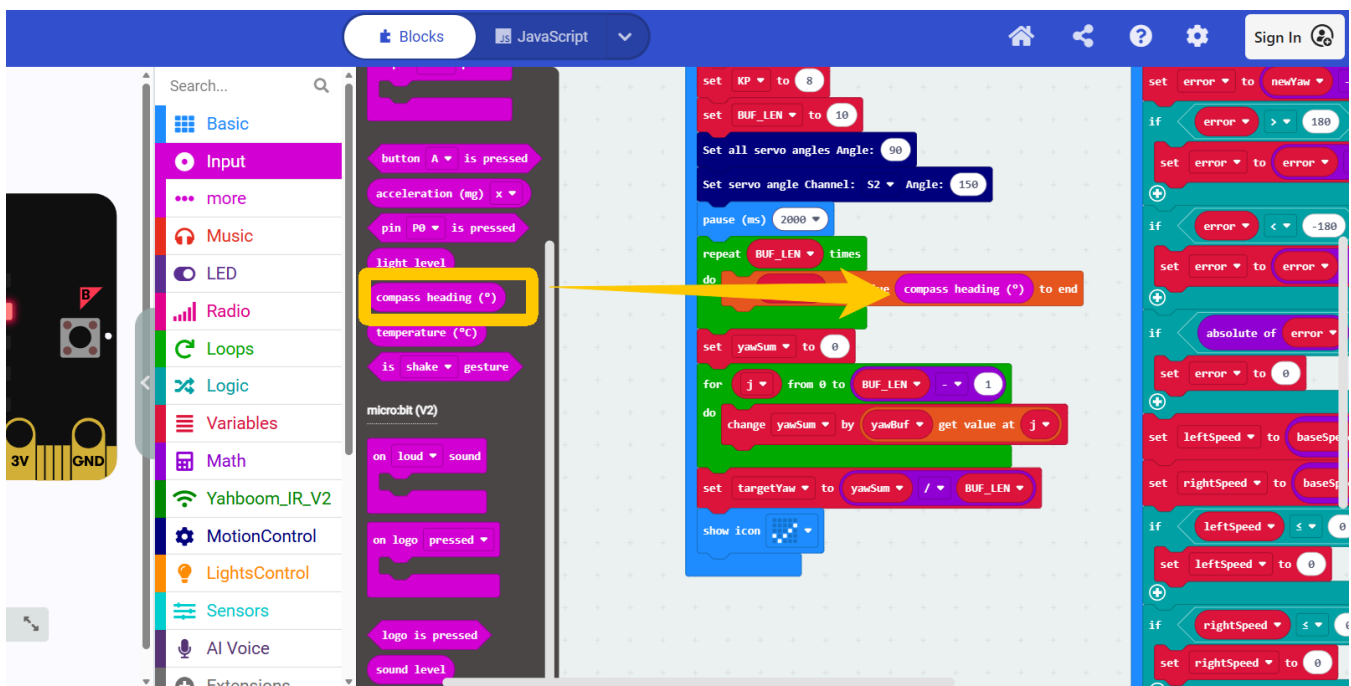
2. Implementation Principle



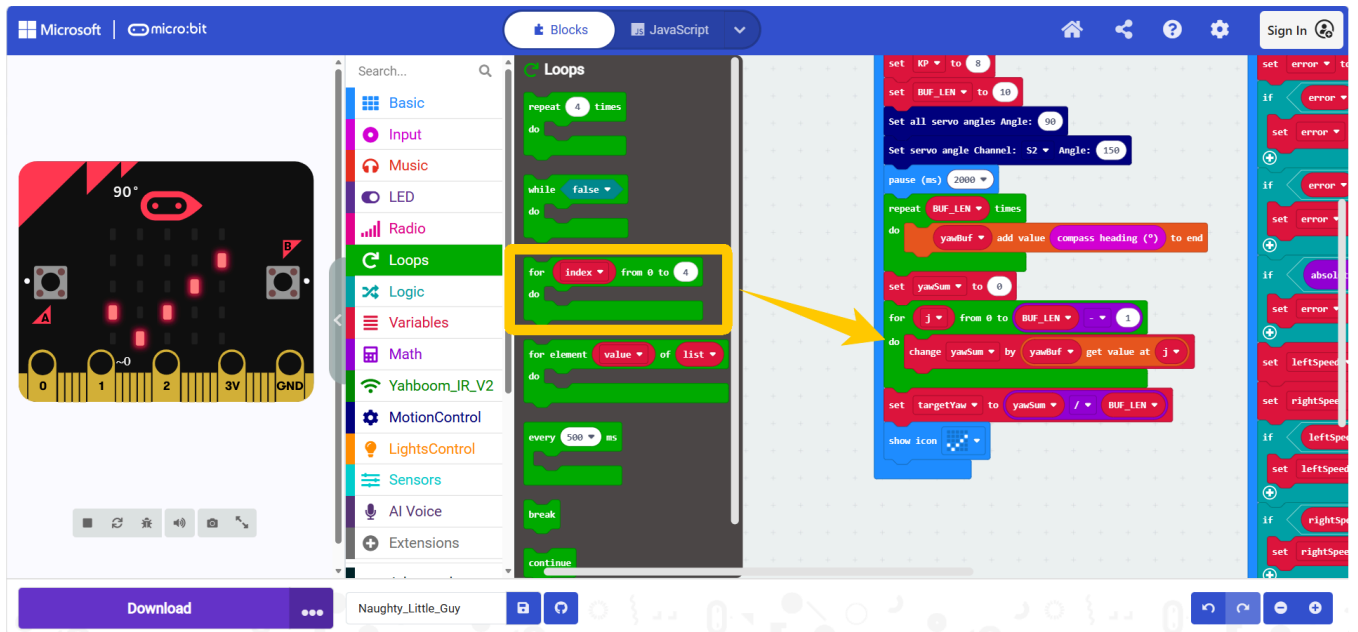
3. Experiment Source Code



Reading the compass heading is located in the **Input** category.



For `j` from 0 to `BUF_LEN-1` is located in the **Loops** category.



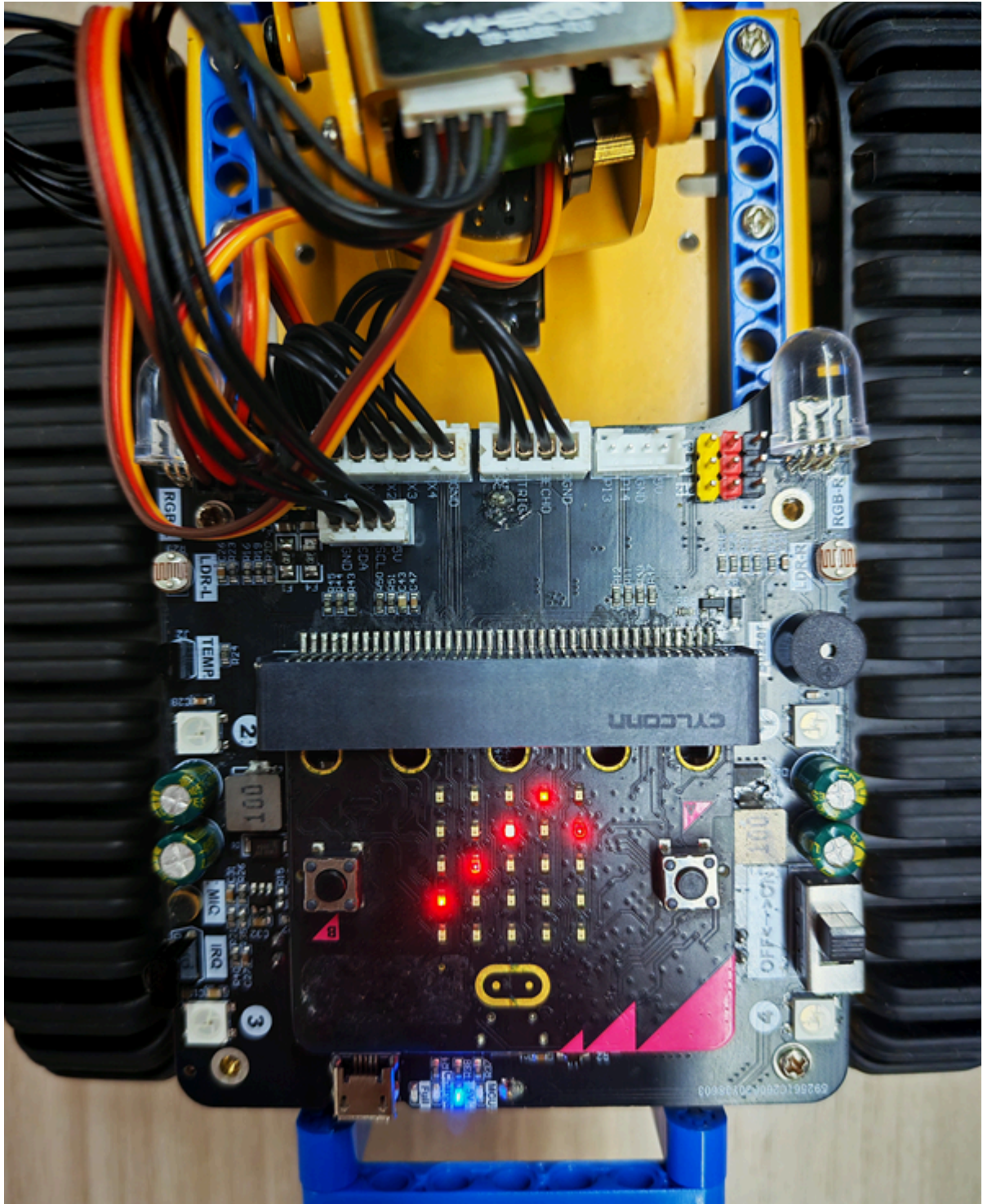
The positions of other blocks have already been introduced in previous tutorials, so we will not go into too much detail here.

4. Experiment Steps

1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable to the Micro:bit. Note: not the Charging interface on the expansion board.
2. Flash `9.Mischievous_Little_Guy.hex` to the Micro:bit. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming".
3. Turn on the car switch (switch it to ON).

5. Experiment Phenomenon

After flashing and turning on the power, the car will first detect the current direction information and display a check mark ✓ on the LED matrix, and then drive straight forward. If an external force rotates the car to the left, the car will turn right; if the car is rotated to the right, the car will turn left.



6. Precautions

If the screen scrolls a message saying "TILT TO FILL SCREEN", this is a warning message from the Microbit, meaning to tilt the device to fill the LED screen. At this time, you need to pick up the car and tilt it in different directions until all its 5x5 LED dots light up, completing the sensor calibration. After calibration is complete, a smiley face will be displayed, and then the car can move normally.